Motion and Ascent

Why orbit is mostly sideways speed, what gravity and drag cost, and how a launch vehicle bends its path from vertical to horizontal.

Isagi · November 2026 · Theory from NASA Glenn's Beginner's Guide to Rockets; ISRO figures checked 9 November 2026

HOW TO USE THIS CHAPTER

- **Print single-sided** and write in the margins.
- Do each example with a calculator first. Use R_E = 6,371 km, μ = 398,600 km³/s², g_0 = 9.81 m/s².
- ISRO publishes event times and orbit for LVM3, not the altitude and speed at each moment. Where I use typical values or assumptions I say so.

By the end of this chapter you can: compute orbital speed and period; explain why a satellite is not "beyond gravity"; estimate gravity loss and drag loss; and describe the three phases of an ascent with LVM3's timeline.

1 What "being in orbit" means

NASA's guide treats flight as the result of forces. Orbit is the same idea with one twist. A satellite is always falling towards the Earth, but it is moving sideways so fast that the ground curves away beneath it as quickly as it falls. At 180 km up, gravity is still 95% as strong as on the ground. Nothing has switched gravity off. The satellite is simply too fast to hit the Earth.

For a circular orbit, gravity supplies exactly the sideways pull needed to bend the path into a circle, which gives:

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v  =  √( μ / r )          period  T  =  2π √( r³ / μ )
r = R_E + altitude,   μ = 398,600 km³/s²
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ISRO LENS: WHY LAUNCH EAST

The Earth already turns eastward. At Sriharikota (about 13.7° N), the ground moves at about 452 m/s, from 465 m/s at the equator times cos(13.7°). A launch to the east starts with that speed for free, which is about 6% of orbital speed. Most Indian launches therefore head east or south-east over the sea; polar missions go south and give up the bonus.

2 The shape of an ascent

A rocket must first climb out of the thickest air, then turn and build up sideways speed. NASA's trajectory discussion divides this into steps, which we group into three phases.

- **Vertical rise.** A few seconds straight up to clear the pad and tower. Fast to fly, but each second straight up spends propellant just holding the rocket against gravity.
- **Pitch-over and gravity turn.** The vehicle tips over gently. After that, gravity itself bends the path towards the horizontal while the engines keep pushing along the flight direction, so there is almost no sideways load on the structure.
- **Horizontal acceleration.** Above the atmosphere, the upper stage burns almost horizontally until it reaches orbital speed.

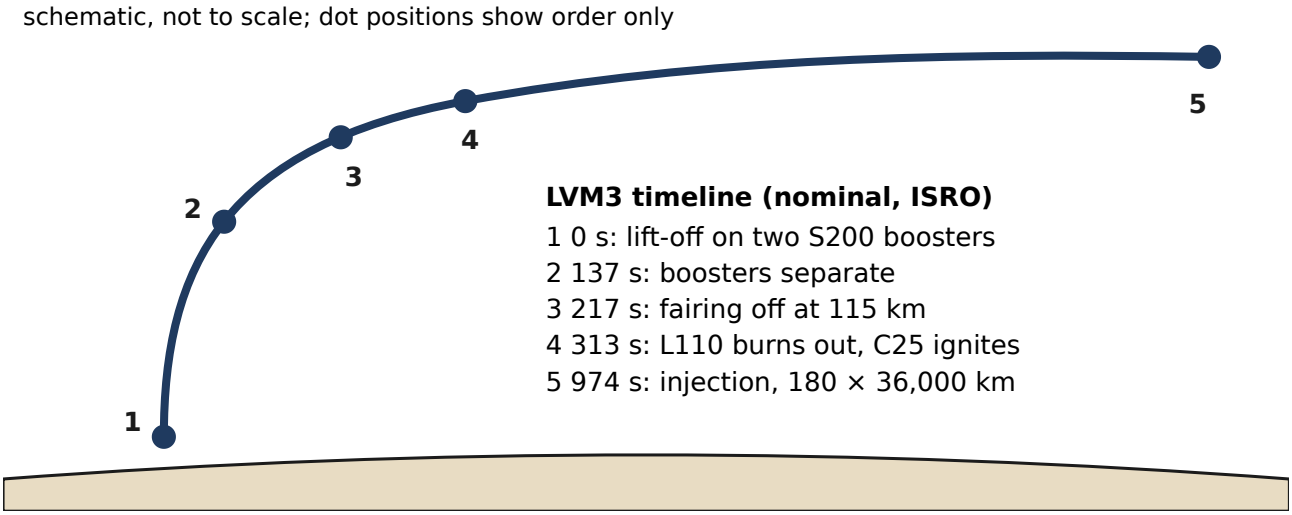


Figure 1. The path bends from vertical to nearly horizontal. Most of the velocity is added in the long, nearly flat final burn.

3 Orbit is mostly speed, not height

Energy per kilogram makes the point. Raising a kilogram to 180 km costs a little potential energy. Giving it 7.8 km/s costs a great deal of kinetic energy.

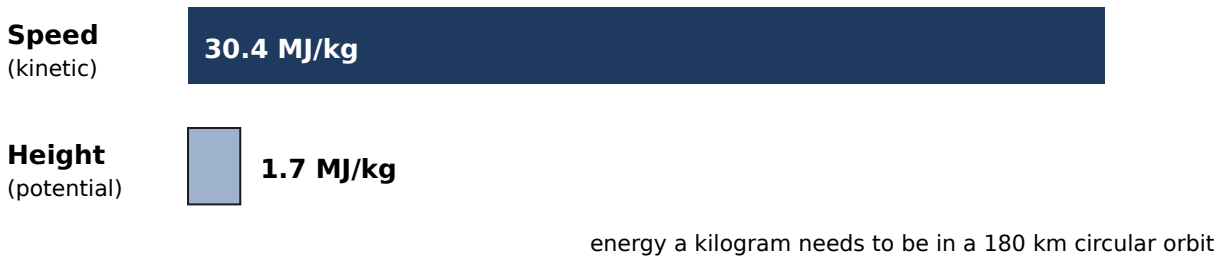


Figure 2. About 95% of the energy is speed. Scale: 15 px per MJ/kg. Worked in Example B.

4 Losses: gravity and drag

Chapter 2's rocket equation gives the Δv the engines add. Not all of it ends up as orbital speed. NASA names the forces that take some away.

Gravity loss. While the rocket climbs, gravity pulls it back at about 9.8 m/s². Every second of burning spends 9.8 m/s of Δv fighting it, whether or not the rocket is gaining useful speed. The worst case is a purely vertical burn, where the loss is g × t. Turning early and burning more horizontally cuts the loss, which is why the gravity turn exists. It also explains why high T/W helps: a short burn loses less.

Drag loss. Drag is ½ × ρ × v^2 × C_d × A, with ρ the air density, v the speed, C_d a shape number and A the frontal area. The quantity ½ρv^2 is called **dynamic pressure, q**. Early on the rocket is slow but the air is dense; later it is fast but the air is nearly gone. In between, q peaks at **max-Q**, the moment of greatest aerodynamic stress. Drag is a small loss for a large rocket, but it sets the structure's strength requirement.

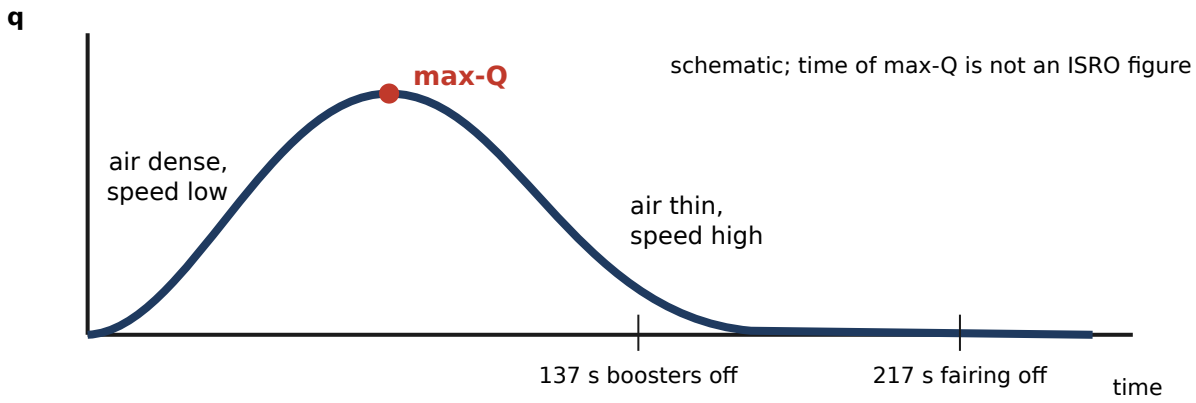


Figure 3. Dynamic pressure rises, peaks, then collapses as the air thins. The fairing stays on until q is low (Chapter 0).

Putting it together, a typical Δv budget to low Earth orbit (public textbook ranges, not ISRO data) is:

Item	Δv (km/s)
Orbital speed at ~180 km	7.8
Gravity loss	about +1.0 to 1.5
Drag and steering loss	about +0.2 to 0.3
Earth’s rotation (eastward launch)	about −0.4
Total needed from the engines	about 8.6 to 9.2, and mission planners add margin

That is why Chapter 0 said “roughly 9 to 10 km/s”: the ideal figure plus real-world margin.

5 Worked examples

Example A: orbital speed and period

```
r = 6,371 + 180 = 6,551 km
v = √(398,600 / 6,551) = √60.84 ≈ 7.80 km/s
T = 2π √(6,551³ / 398,600) ≈ 5,277 s ≈ 88 minutes
At 400 km (ISS height): v ≈ 7.67 km/s
```

Higher orbits move slower. Lower orbits are faster and circle the Earth more often.

Example B: energy of an orbit

```
Kinetic = ½ v² = ½ × (7,800)²            ≈ 30.4 MJ/kg
Potential = μ (1/R_E − 1/r)
   = 3.986×10¹⁴ × (1/6.371×10⁶ − 1/6.551×10⁶)
   ≈ 1.72 MJ/kg
Speed share = 30.4 / (30.4 + 1.72) ≈ 95%
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The hard part of reaching orbit is not height. It is speed.

Example C: gravity loss

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Worst case (vertical burn):  Δv loss = g × t
S200 burn, 134 s:    9.81 × 134 ≈ 1,315 m/s
g at 41.5 km (L110 ignition) = 9.81 × (6371/6412.5)² ≈ 9.68 m/s²
g at 115 km (fairing off)    = 9.81 × (6371/6486)²   ≈ 9.46 m/s²
```

A real ascent loses less than 1.3 km/s in the booster phase, because the path tilts and the vehicle also gains upward speed. Gravity also weakens only slowly with height: it is still 94% of surface value at 115 km.

Example D: what the transfer orbit asks of LVM3

The target is a 180 × 36,000 km orbit. Use the vis-viva form $v = \sqrt{\mu(2/r - 1/a)}$, with $a = (6{,}551 + 42{,}371)/2 = 24{,}461$ km.

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Perigee (r = 6,551 km):
   v = √(398,600 × (2/6,551 − 1/24,461)) ≈ 10.27 km/s
Apogee (r = 42,371 km):
   v = √(398,600 × (2/42,371 − 1/24,461)) ≈ 1.59 km/s
Period = 2π √(24,461³ / 398,600) ≈ 38,070 s ≈ 10.6 h
Circular speed at apogee: √(398,600 / 42,371) ≈ 3.07 km/s
Extra speed to circularise: 3.07 − 1.59 ≈ 1.48 km/s
```

LVM3 gives the satellite 10.27 km/s at perigee; coasting, it slows to 1.59 km/s at apogee. The last 1.48 km/s is added later by the satellite's own engine, so ISRO's job ends at the transfer orbit.

Example E: how big is max-Q? (a what-if)

Assumed values, not ISRO data: altitude 11 km (air density 0.364 kg/m³), speed 400 m/s, $C_d = 0.3$, frontal area from the 4 m body, $A = \pi \times 2^2 = 12.6$ m², vehicle mass 400 t.

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q = ½ × 0.364 × 400² ≈ 29,100 Pa ≈ 29 kPa
Drag = C_d × q × A = 0.3 × 29,100 × 12.6 ≈ 110 kN
Weight = 400,000 × 9.81 ≈ 3.9 MN      Drag / weight ≈ 2.8%
```

Drag is a small fraction of the weight, yet 29 kPa of dynamic pressure across the whole nose is what the fairing and body must be built to survive.

CHAPTER SUMMARY

- Orbit means falling around the Earth: $v = \sqrt{(\mu/r)} \approx 7.8$ km/s at 180 km, period 88 min. Gravity is still about 95% of its surface value there.
- About 95% of orbital energy is speed. The ascent turns from vertical to horizontal to build it.
- Losses: gravity (up to $g \times t$ for a vertical burn) and drag (peaks at max-Q); Earth's rotation helps eastward launches by up to about 0.45 km/s.
- Typical Δv to low orbit is about 8.6 to 9.2 km/s plus margin.
- LVM3's job ends at the 180 × 36,000 km transfer orbit (10.27 km/s at perigee, 1.59 km/s at apogee).

Next, Chapter 4: aerodynamics. Where drag comes from, why shape matters and how fast air changes the rules.

Practice problems

R_E = 6,371 km, μ = 398,600 km³/s², g = 9.81 m/s². Answers are on the last page.

P1. Find the speed and period of a circular orbit at 300 km.

P2. By what percentage is gravity weaker at 400 km than at the surface?

P3. The L110 burns for 200 s. What is the maximum gravity loss if it burned vertically the whole time? Why is the real loss smaller?

P4. Find the kinetic energy per kilogram of the 300 km orbit in P1.

P5. Dynamic pressure at sea level (ρ = 1.225 kg/m³) at 250 m/s: compute q and compare with Example E’s 29 kPa. What does it tell you about when q is highest?

P6. The ground at Sriharikota moves east at 452 m/s. How much speed difference is there between launching due east and due west? Which saves propellant?

Glossary

Orbit
A path where sideways speed and gravity balance, so the object keeps falling around the Earth.

Orbital speed
$\sqrt{(\mu/r)}$ for a circular orbit.

Period
Time for one full orbit.

Gravity turn
A path that gravity itself bends towards the horizontal while thrust stays along the flight direction.

Gravity loss
Δv spent holding the rocket against gravity while it climbs.

Drag loss
Δv spent against air resistance.

Dynamic pressure (q)
$\frac{1}{2}\rho v^2$, the air's push on the vehicle.

Max-Q
The moment of peak dynamic pressure.

Perigee, apogee
Lowest and highest points of an elliptical orbit.

Transfer orbit (GTO)
An elliptical orbit that reaches geosynchronous height at its apogee.

Answer key

Fold this page back before you start the problems.

P1. r = 6,671 km. v = √(398,600/6,671) ≈ 7.73 km/s. T = 2π√(6,671³/398,600) ≈ 5,422 s ≈ 90.4 min.

P2. (6,371/6,771)² = 0.885, so gravity is 11.5% weaker (still 88.5% of surface value).

P3. 9.81 × 200 ≈ 1,962 m/s. Real loss is smaller because the path tilts, so gravity opposes only part of the motion along the flight path.

P4. ½ × (7,730)² ≈ 29.9 MJ/kg.

P5. q = ½ × 1.225 × 250² ≈ 38,300 Pa ≈ 38 kPa. That is more than Example E's 29 kPa even though the speed is lower: dense air matters more than speed alone. Real peak q occurs where rising speed and falling density balance, which is why max-Q comes soon after lift-off.

P6. East gains 452 m/s, west loses 452 m/s, a difference of 904 m/s (about 11.6% of 7.8 km/s). Launching east saves propellant.

Sources

Theory: NASA Glenn Research Center, Beginner's Guide to Rockets (forces, trajectory, drag). ISRO: LVM3 launch vehicle page (event timeline, transfer orbit, fairing separation altitude). Constants (R_E, μ, Earth rotation, Sriharikota latitude) and the Δv-budget ranges are standard textbook and public values; Example E uses assumed numbers, not ISRO data.